

MATH 2101 LINEAR ALGEBRA I
FALL SEMESTER 2025
TEST 1 – OUTLINE OF SOLUTION

- (1) (4 points) Find the inverse of the following matrix using *adjugate matrix*:

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & 1 \\ 0 & 2 & -1 \end{pmatrix}.$$

Sol. By direct computation, we have (let A be the matrix above)

$$\text{adj}(A) = \begin{pmatrix} -2 & -1 & -2 \\ 2 & -1 & -2 \\ 2 & -1 & 2 \end{pmatrix}^T = \begin{pmatrix} -2 & 2 & 2 \\ -1 & -1 & -1 \\ -2 & -2 & 2 \end{pmatrix}.$$

The determinant is $\det(A) = -4$. So the inverse of A is

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = -\frac{1}{4} \begin{pmatrix} -2 & 2 & 2 \\ -1 & -1 & -1 \\ -2 & -2 & 2 \end{pmatrix}.$$

- (2) (6 points) Let V be a vector space. Determine whether the following statements are true or false. Briefly explain your answer.

- (a) If S is a linearly dependent subset in V , then each vector in S is a linear combination of other vectors in S .
- (b) Subsets of linearly dependent sets in V are also linearly dependent.

Sol.

- (a) False. Counterexample: Take $V = \mathbb{R}^2$, $S = \{(1, 1)^T, (2, 2)^T, (1, 2)^T\}$, note that S is a linearly dependent subset of V , but $(1, 2)^T$ is not in the span of the other two vectors.
- (b) False. Consider the example above, we observe that $(1, 1)^T$ and $(1, 2)^T$ are linearly independent.

- (3) (4 points) Let V be the vector space of functions from $\mathbb{R}$ to $\mathbb{R}$. Define

$$S = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f(1) = 1\}.$$

Determine whether S forms a subspace of V . Explain your answer.

Sol. The set S is not a vector subspace of V . One can see this by showing S is not closed under addition. More precisely, pick $f, g \in S$, we note that

$$(f + g)(1) = f(1) + g(1) = 2 \neq 1.$$

One can also disprove by showing S does not have an additive identity (which is the zero function in V).

- (4) (5 points) Given the following system of linear equations

$$2x_1 + 3x_2 + ax_3 = 1$$

$$4x_1 + 6x_2 + bx_3 = 2$$

What can one say about the number of solutions for the system for different a and b ? Explain your answer.

Sol. We write the system in terms of matrices:

$$\begin{pmatrix} 2 & 3 & a \\ 4 & 6 & b \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The associated augmented matrix reads

$$\left(\begin{array}{ccc|c} 2 & 3 & a & 1 \\ 4 & 6 & b & 2 \end{array} \right).$$

By applying row operations on the above augmented matrix, we have

$$\left(\begin{array}{ccc|c} 1 & 3/2 & a/2 & 1/2 \\ 0 & 0 & b-2a & 0 \end{array} \right).$$

If $b - 2a \neq 0$, then only x_2 is a free variable and $x_3 = 0$. As a result, there are infinitely many solutions. On the other hand, if $b - 2a = 0$, the solution set becomes nothing but the solution set for $x_1 + \frac{3}{2}x_2 + \frac{a}{2}x_3 = \frac{1}{2}$, which contains infinitely many elements. Therefore, for any a, b , the system has infinitely many solutions.

- (5) (6 points) Let A be an $n \times m$ matrix.

(a) Suppose $n = m$ and A is invertible. Find the reduced row echelon form of

$$\begin{pmatrix} A & B \\ 0 & A \end{pmatrix}$$

for some $n \times m$ matrix B . Explain your answer.

(b) Let r be the number of leading ones in the RREF of A . What is the maximum possible number of leading ones in the reduced row echelon form of

$$\begin{pmatrix} A & B \\ 0 & A \end{pmatrix}.$$

Sol.

- (a) By a similar proof as in assignment 2, problem 1, we know that the indicated matrix is invertible. So the RREF is I_{2n} .
- (b) By assumption, we know $0 \leq r \leq \min\{m, n\}$. The number of leading ones in the RREF corresponds to the rank of the block matrix, i.e., the dimension of the column space. Note that it is the sum of $\text{rank}(A)$ and $\text{rank}\begin{pmatrix} A & B \end{pmatrix}$, where the first one is r and the latter is at most $\min\{m, n\}$. The equality can be obtained by choosing a full rank $n \times m$ matrix for B . So the maximum possible number of leading ones in the RREF is $r + \min\{m, n\}$.